

Apportioning Causal Responsibility of Two Risk Factors for an Adverse Outcome via Counterfactual Attribution

Shanshan Luo

Beijing Technology and Business University

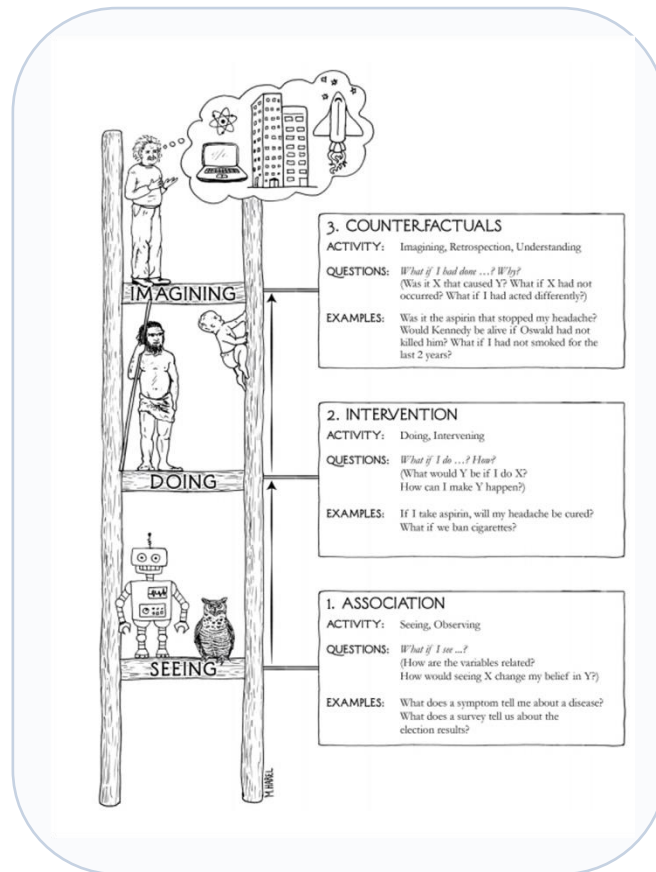
Young Scholars Workshop (JCSDS 2026)



arXiv:2606.18459

What is causal attribution analysis?

Forward-looking analysis studies the effects of causes — what would happen;
attribution analysis studies the causes of effects — why it happened.



1

Continuous outcome attribution

Causal attribution analysis for continuous outcomes.

Shanshan Luo, Yixuan Yu, Chunchen Liu, Feng Xie, and Zhi Geng.

ICML, Vancouver, Canada, 2025. (spotlight)

2

Interactive causes of an occurred outcome

Assessing interactive causes of an occurred outcome due to two binary exposures.

Shanshan Luo, Wei Li, Xueli Wang, Shaojie Wei, and Zhi Geng.

To appear in Biometrics, 2026.

3

Responsibility apportionment

Apportioning causal responsibility of two risk factors for an adverse outcome via counterfactual attribution.

Shanshan Luo, Yafang Deng, Qingyuan Zhao, and Zhi Geng.

arXiv, 2026.

Collaborators



Yafang Deng

Second-year Master Student

School of Mathematics and Statistics
Beijing Technology and Business University



Qingyuan Zhao

Professor

Statistical Laboratory
University of Cambridge



Zhi Geng

Professor

School of Mathematical Sciences
Peking University
Beijing Technology and Business University

The attribution problem: why did Bob get lung cancer?

Bob's claim



observed case:
smoking + asbestos
+ lung cancer

Total loss: £10,000

Bob seeks compensation.
The court asks how to split
£10,000.

Lung cancer rates



$\xi_{00} = 0.12\%$



$\xi_{10} = 0.95\%$



$\xi_{01} = 0.67\%$



$\xi_{11} = 4.51\%$

👉 Smoking's causal effect: $\xi_{10} - \xi_{00} = 0.83\%$

👉 Asbestos's causal effect: $\xi_{01} - \xi_{00} = 0.55\%$

The court asks: how should responsibility be apportioned between smoking and asbestos? Who should pay what share?

Smoking company

Did smoking contribute
to this cancer?

Asbestos / employer

Did workplace asbestos
contribute?

Background risk

Would it have happened
anyway?

Fair responsibility rule

?

Only one mechanism is true for Bob, but we do not observe it.

Story	Counterfactual pattern (worlds with cancer)	Responsibility
Synergy	   	Smoking: 50% Asbestos: 50%
Smoking only	   	Smoking: 100% Asbestos: 0%
Asbestos only	   	Smoking: 0% Asbestos: 100%
Parallel	   	Smoking: 50% Asbestos: 50%
Background	   	Smoking: 0% Asbestos: 0%

So we put statistical weights on the stories...

Closed-form posterior responsibilities

v is an **unidentified** parameter defined in terms of two counterfactual worlds.

Story	Counterfactual pattern (worlds with cancer)	Fair Responsibility Rule	Statistics Weight (randomization & monotonicity)
Synergy		Smoking: 50% Asbestos: 50%	$\frac{\xi_{11} - \xi_{10} - \xi_{01} + v}{\xi_{11}}$
Smoking only		Smoking: 100% Asbestos: 0%	$\frac{\xi_{01} - v}{\xi_{11}}$
Asbestos only		Smoking: 0% Asbestos: 100%	$\frac{\xi_{10} - v}{\xi_{11}}$
Parallel		Smoking: 50% Asbestos: 50%	$\frac{v - \xi_{00}}{\xi_{11}}$
Background		Smoking: 0% Asbestos: 0%	$\frac{\xi_{00}}{\xi_{11}}$

Smoking's responsibility:
$$\frac{\xi_{11} + \xi_{10} - \xi_{01} - \xi_{00}}{2\xi_{11}}$$

Asbestos's responsibility:
$$\frac{\xi_{11} + \xi_{01} - \xi_{10} - \xi_{00}}{2\xi_{11}}$$

Numerical result: a compensation story

A £10,000 loss is translated into responsibility shares for the occurred cancer.

Observed risks: ξ_{00} =0.12% ξ_{10} =0.95% ξ_{01} =0.67% ξ_{11} =4.51%

Smoking’s responsibility: $\frac{\xi_{11}+\xi_{10}-\xi_{01}-\xi_{00}}{2\xi_{11}} = 51.77\%$,

Asbestos’s responsibility: $\frac{\xi_{11}+\xi_{01}-\xi_{10}-\xi_{00}}{2\xi_{11}} = 45.57\%$.

Bob's claim



Observed case:
smoking + asbestos
+ lung cancer

Total loss: £10,000

Bob seeks compensation.
The court asks how to split
£10,000.

Who should pay what
share?



Smoking / tobacco company

51.77%

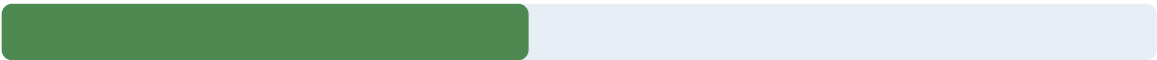
£5,177



Workplace asbestos / employer

45.57%

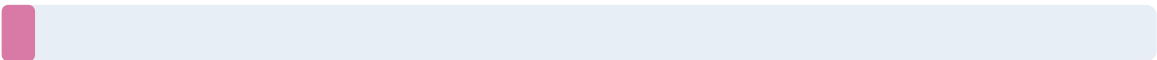
£4,557



Background vulnerability

2.66%

£266



Interpretation: smoking bears slightly more than half; asbestos bears almost the other half; background risk is small but not zero.

Takeaways

1

Responsibility is a counterfactual claim

We ask whether smoking or asbestos mattered for this occurred cancer case, not just whether they raise average risk.

2

Mechanism stories come before percentages

Smoking and asbestos may act jointly, separately, in parallel, or not at all for this individual.

3

Average responsibility is identifiable

Under monotonicity and a balanced rule, the posterior average responsibility is determined by observable risks.



arXiv:2606.18459

Thank you!